

Residual-Guided Hidden-Physics Discovery for PFM-Constrained BTO Phase-Field Simulations

sim_bo / AutoPF–MatEnsemble campaign summary

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1 Purpose and Data Products

The latest hidden-physics campaign was designed to answer a focused question: after fixing the gradient coefficients to the dense-posterior, physically tangible values

$$g_{11} = 0.5, \quad g_{12} = -0.06, \quad g_{44} = 0.02,$$

what additional low-dimensional physics terms most consistently reduce the PFM surface-displacement residuals?

The current best model is a co-active surface-screening field plus a localized anisotropic, shear-like eigenstrain component. The final full-holdout round is `round_anisotropic_holdout_full_000`. Its analysis files are under

`sim_bo_bto_physical_bounds/anisotropic_hidden_analysis.`

The MOOSE input used for the hidden-physics model is `BTO_DW_anisotropic_hidden_physics_zdecay_ic.i`; the campaign manifests were produced by `build_anisotropic_hidden_physics_manifest.py` and `build_anisotropic_holdout_validation_manifest.py`.

2 Observation Model and Residual Objective

For each experimental bias–pulse condition

$$c = (V, \tau),$$

the experiment provides a surface PFM-derived observable, denoted $u_{z,c}^{\text{exp}}(x, y)$. The phase-field simulation gives a final surface displacement $u_{z,c}^{\text{sim}}(x, y; \theta)$, where θ collects the gradient coefficients and hidden-physics parameters.

Before evaluating the objective, the simulated surface field is transposed into the experimental orientation and resampled to the experimental grid. Both fields are then z-normalized condition-by-condition:

$$\hat{u}_{z,c}^{\text{sim}} = \frac{u_{z,c}^{\text{sim}} - \langle u_{z,c}^{\text{sim}} \rangle}{\text{std}(u_{z,c}^{\text{sim}})}, \quad \hat{u}_{z,c}^{\text{exp}} = \frac{u_{z,c}^{\text{exp}} - \langle u_{z,c}^{\text{exp}} \rangle}{\text{std}(u_{z,c}^{\text{exp}})}.$$

The residual map is

$$R_c(x, y; \theta) = \hat{u}_{z,c}^{\text{sim}}(x, y; \theta) - \hat{u}_{z,c}^{\text{exp}}(x, y).$$

The objective uses the center crop, excluding the outer 10% edge band in each image dimension. If $M(x, y) \in \{0, 1\}$ is the crop mask, the normalized MSE is

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{\sum_{x,y} M(x, y) R_c(x, y; \theta)^2}{\sum_{x,y} M(x, y) (\hat{u}_{z,c}^{\text{exp}}(x, y))^2}.$$

The edge masking was important because the largest residuals in the baseline fits were often dominated by lateral/edge artifacts rather than by the domain interior that should drive hidden-physics inference.

3 Baseline Physics Model

The core phase-field model is an LGD electro-elasto-static model with polarization $\mathbf{P} = (P_x, P_y, P_z)$, displacement $\mathbf{u}$, and electric potential ϕ . Schematically, the relaxed state minimizes or evolves down the functional

$$\mathcal{F}_0 = \int_{\Omega} [f_{\text{Landau}}(\mathbf{P}) + f_{\text{grad}}(\nabla \mathbf{P}) + f_{\text{elec}}(\mathbf{P}, \phi) + f_{\text{el}}(\boldsymbol{\varepsilon}, \mathbf{P})] \, d\Omega.$$

The gradient term is controlled by the three cubic coefficients (g_{11}, g_{12}, g_{44}) . In compact tensor notation,

$$f_{\text{grad}} = \frac{1}{2} G_{ijkl} \frac{\partial P_i}{\partial x_j} \frac{\partial P_k}{\partial x_l},$$

where cubic symmetry reduces G_{ijkl} to the three fitted coefficients.

Mechanical equilibrium is solved with small strain

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

and the elastic energy is

$$f_{\text{el}} = \frac{1}{2} C_{ijkl} \varepsilon_{ij}^e \varepsilon_{kl}^e.$$

The elastic strain includes electrostrictive and hidden eigenstrain components:

$$\varepsilon_{ij}^e = \varepsilon_{ij} - \varepsilon_{ij}^Q(\mathbf{P}) - \varepsilon_{ij}^{\text{iso}}(\mathbf{x}) - \varepsilon_{ij}^{\text{aniso}}(\mathbf{x}),$$

with electrostriction represented schematically as

$$\varepsilon_{ij}^Q = Q_{ijkl} P_k P_l.$$

The hidden-off baseline has

$$\varepsilon_{ij}^{\text{iso}} = 0, \quad \varepsilon_{ij}^{\text{aniso}} = 0, \quad \lambda_{\text{scr}} = 0, \quad h_{\text{flexo}} = 0.$$

4 Hidden-Physics Parameterizations

All hidden-physics candidates were deliberately low-dimensional. The goal was not to fit a pixelwise field by BO, but to test physically interpretable families that could explain structured residuals.

4.1 Surface Screening / Depolarization Proxy

The surface-screening field is represented by a localized Gaussian envelope:

$$\lambda_{\text{scr}}(\mathbf{x}) = \lambda_0 + A_{\text{scr}} \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma_{xy,\text{scr}}^2} \right] \exp \left[-\frac{(z - z_0)^2}{2\sigma_{z,\text{scr}}^2} \right].$$

In the latest holdout model,

$$A_{\text{scr}} = 0.08, \quad \sigma_{xy,\text{scr}} = 80 \text{ nm}, \quad \sigma_{z,\text{scr}} = 25 \text{ nm}.$$

In the MOOSE input this appears as `spatial_screen_lambda` and enters the P_z equation through `screening_depolar_z`. The residual-level implementation is proportional to

$$R_{P_z}^{\text{scr}} \propto -\frac{1}{2\epsilon_b} \lambda_{\text{scr}}(\mathbf{x}) \langle P_z \rangle,$$

with $\epsilon_b = \text{screen_permittivity} = 0.08854187$. This term should be read as an effective depolarization/screening proxy rather than a complete microscopic charge-transport model.

4.2 Isotropic Vegard-Like Eigenstrain

The isotropic Vegard-like field was included as

$$\varepsilon_{ij}^{\text{iso}}(\mathbf{x}) = \eta_{\text{iso}}(\mathbf{x}) \delta_{ij},$$

where

$$\eta_{\text{iso}}(\mathbf{x}) = \eta_0 + A_{\text{iso}} \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma_{xy,\text{iso}}^2} \right] \exp \left[-\frac{(z - z_0)^2}{2\sigma_{z,\text{iso}}^2} \right].$$

In the final anisotropic holdout winner the isotropic Vegard amplitudes are zero. Thus the final interpretation is not a uniform volumetric vacancy expansion/contraction; it is a more directional elastic signature.

4.3 Low-Rank Anisotropic Eigenstrain

The anisotropic hidden field is a rank-restricted symmetric eigenstrain tensor:

$$\varepsilon^{\text{aniso}}(\mathbf{x}) = \Phi_A(\mathbf{x}) \mathbf{A},$$

where

$$\Phi_A(\mathbf{x}) = \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma_{xy,A}^2} \right] \exp \left[-\frac{(z - z_0)^2}{2\sigma_{z,A}^2} \right],$$

and

$$\mathbf{A} = \begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{xy} & A_{yy} & A_{yz} \\ A_{xz} & A_{yz} & A_{zz} \end{pmatrix}.$$

The MOOSE implementation uses separate `ComputeEigenstrain` blocks for xx, yy, zz, xy, xz, yz and adds all of them to `eigenstrain_names` in the small-strain material.

The best final candidate has only the yz component active:

$$A_{yz} = 7.5 \times 10^{-4}, \quad A_{xx} = A_{yy} = A_{zz} = A_{xy} = A_{xz} = 0,$$

with

$$\sigma_{xy,A} = 120 \text{ nm}, \quad \sigma_{z,A} = 35 \text{ nm}.$$

This is a localized shear-like eigenstrain. A physically cautious interpretation is that the residual field is consistent with asymmetric defect/dislocation-mediated elastic accommodation, rather than with a purely isotropic vacancy Vegard strain.

4.4 Phenomenological Flexo Proxy

A full flexoelectric model would include Lifshitz-type couplings between polarization gradients and strain gradients, for example terms of the form

$$f_{\text{flexo}} \sim \mu_{ijkl} \left(P_i \frac{\partial \varepsilon_{jk}}{\partial x_l} - \varepsilon_{jk} \frac{\partial P_i}{\partial x_l} \right),$$

which generally requires additional carefully verified kernels/materials. The present campaign did not implement that rigorous tensorial flexoelectric energy. Instead, it tested a phenomenological internal-field proxy acting on P_z :

$$h_{\text{flexo}}(\mathbf{x}) = \left[a_0 + a_1 \frac{z - z_0}{\sigma_{z,f}} \right] \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma_{xy,f}^2} \right] \exp \left[-\frac{(z - z_0)^2}{2\sigma_{z,f}^2} \right].$$

Equivalently, one may view the proxy as an energy contribution

$$\mathcal{F}_{\text{proxy}} = - \int_{\Omega} h_{\text{flexo}}(\mathbf{x}) P_z(\mathbf{x}) \, d\Omega,$$

whose variational derivative drives the P_z equation. In MOOSE this is the `flexo_proxy_pz` body force with function `flexo_proxy_drive`. The best flexo-proxy competitor used

$$a_1 = -10^{-3}, \quad a_0 = 0,$$

and reached a full-holdout nMSE of 0.0687001, slightly worse than the anisotropic shear-like eigenstrain.

5 Candidate Ranking and Quantitative Results

The full 30-condition holdout comparison is:

rank	candidate	active terms	mean nMSE
1	<code>shear_yz_pos</code>	$\lambda_{\text{scr}} + A_{yz}\Phi_A$	0.0660709
2	<code>flexo_proxy_grad_neg</code>	$\lambda_{\text{scr}} + h_{\text{flexo}}$	0.0687001
3	<code>screening_only_refined_control</code>	λ_{scr}	0.0698688
4	<code>fixed_g_hidden_off</code>	none	0.0977860

The best anisotropic candidate reduces nMSE by

$$\frac{0.0977860 - 0.0660709}{0.0977860} \times 100 = 32.43\%$$

relative to the hidden-off fixed- g model. It improves all 30/30 holdout conditions relative to hidden-off, 23/30 relative to the screening-only control, and 21/30 relative to the flexo-proxy competitor.

6 Scientific Interpretation

The important point is not only that the residual decreases, but that the residual decreases after each candidate is run through the MOOSE phase-field solver. Thus the optimization is physics-constrained in the practical sense: the learned correction must survive coupled LGD, electrostatic, and mechanical relaxation. The loss does not currently include an explicit free-energy penalty; free energy is used diagnostically.

The current best hidden physics is therefore best described as:

localized screening + localized anisotropic ε_{yz}^* eigenstrain.

This is consistent with a sample containing defects/dislocations that produce directional elastic accommodation near the switched/nucleated region. It is not, by itself, proof of a unique microscopic defect population. The anisotropic eigenstrain should be presented as an inferred effective field whose symmetry and localization provide a physics cue for follow-up experiments or higher-fidelity defect models.

7 Regenerated Figure Files

The latest manuscript figures were regenerated by `paper_figures/make_latest_campaign_fig3_fig4_story_panels.py`. The requested copies are:

- `paper_figures/needs_regeneration_for_the_latest_campaign/Figure3_hidden_physics_discovery.pdf`
- `paper_figures/needs_regeneration_for_the_latest_campaign/fig4_holdout_validation_redesigned.pdf`

Figure 3a now includes the hidden-off nMSE and intentionally omits error bars. The two holdout residual atlas panels in Figure 3 use the same center-crop z-normalized residual color scale,

$$R_c \in [-1.2624590595, 1.2624590595],$$

so the apparent residual reduction is visually comparable between hidden-off and hidden-on panels.